Today we will use what we discovered last time to talk about <u>standing waves</u> on a string and in an air column. Specifically, we will use:

1) Reflection at a rigid boundary: $R=-1$ and $T=0$
   $\Rightarrow$ $\pi$ phase shift for reflected wave, no transmission

2) Reflection at a loose boundary: $R=1$ and $T=2$
   $\Rightarrow$ no phase shift for reflected wave, no transmission
   since $T_e = \frac{P_t}{P_i} = 0$ despite $T=2$.

Let's first look at waves in a string fixed at both ends:



The string has tension $T$ and linear mass density $\mu$

If a wave is sent travelling to the right (choose $\phi_0=0$)

$$y_i(x,t) = A\cos(kx-\omega t)$$

then there will be a reflected wave at the RH boundary with a $\pi$ phase shift travelling to the left:

$$y_r(x,t) = A\cos(kx + \omega t + \pi)$$

This wave is then reflected again at LH boundary and becomes the incident wave again due to another $\pi$ phase shift.

The total wave is the sum of the two:

$$y(x,t) = A\cos(kx - \omega t) + A\cos(kx + \omega t + \pi)$$
$$= A\cos(kx - \omega t) - A\cos(kx + \omega t)$$

Use trig identity $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$

$$y(x,t) = A(\cos kx \cos\omega t + \sin kx \sin\omega t)$$
$$- A(\cos kx \cos\omega t - \sin kx \sin\omega t)$$

$$\boxed{y(x,t) = (2A\sin kx)\sin\omega t}$$

The result is a standing wave (not visibly moving left or right) with an amplitude profile given by $2A\sin kx$ that oscillates up and down with frequency $\omega$.

We can redefine $2A \rightarrow A$ since $y(x,t)$ represents the total wave and we are really interested in the total amplitude.

$$\Rightarrow \quad y(x,t) = (A \sin kx) \sin \omega t$$

The ends of the string are fixed at $x = 0, L$ so the condition needed for a standing wave is:

$$\sin(k(0)) = 0 \quad \checkmark \text{ automatically satisfied}$$

$$\sin(kL) = 0 \Rightarrow \quad kL = n\pi$$

Now, $k = \frac{2\pi}{\lambda}$ so this condition can be expressed as

$$\frac{2\pi}{\lambda_n} L = n\pi \quad \text{or} \quad \boxed{\lambda_n = \frac{2L}{n}} \qquad n = 1, 2, 3, \ldots$$

As can be seen on the accompanying slide, the number $n$ is the number of antinodes, while $n+1$ gives the number of nodes (including the fixed ends). The <u>resonant</u> modes are the ones for which a half-integer number of wavelengths fit in the length $L$ of the string. This can be re-cast in terms of resonant frequencies via

$$k_n = \frac{\omega_n}{v} = \frac{\omega_n}{\sqrt{T/\mu}} \Rightarrow \boxed{\omega_n = \frac{n\pi}{L}\sqrt{\frac{T}{\mu}}} \text{ or } \boxed{f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$$

The $f_n$ are called <u>harmonics</u> of the string and $f_1$ is called the fundamental harmonic or the fundamental frequency. The number $n$ is called the harmonic number.

5 minute break

⇒ Top Hat Question

What about compression waves in an air column? There are 2 cases to consider here:

1) One end closed, one end open
2) Both ends open

For a closed end we know there has to be a node. This is because the incident and reflected waves are $\pi$ rad out of phase, so they add to zero at the fixed end.

For an open end, there is no phase shift, so the incident and reflected waves add to give a maximum at the open end.

Instead of adding the waves together again like we did for the fixed string, we can use these two conditions to study the resonance behaviour of air columns without a bunch of trig identities.

1) One end closed, one end open

As can be seen on the accompanying slide, the first 3 harmonics satisfy:

$$\lambda_1 = 4L \qquad \lambda_2 = \frac{4}{3}L \qquad \lambda_3 = \frac{4}{5}L$$

$$\Rightarrow \boxed{\lambda_n = \frac{4L}{n}} \qquad n = 1, 3, 5, \ldots$$

Note that the harmonic number in this case is always odd. The fact that one of the ends is open means that all the even harmonics are missing and only the odd harmonics survive. When you see "the third harmonic" for one end open, one closed, it means that n = 3 above in the usual way. It does not mean that it is the third non-zero harmonic, which would be n = 5.

In terms of frequencies $f_n = \frac{v}{\lambda_n}$

$$\boxed{f_n = \frac{nv}{4L}} \qquad n = 1, 3, 5, \ldots$$

2) Both ends closed

As can be seen on the accompanying slide, this case is the opposite of the fixed string when it comes to the placement of nodes and antinodes, so the same

condition applies here:

$$\lambda_n = \frac{2L}{n} \quad n = 1, 2, 3, \ldots \quad \text{and} \quad f_n = \frac{nv}{2L} \quad n = 1, 2, 3, \ldots$$

Example

Why does your voice go higher when you inhale helium? Your voice box is like a tube open at both ends, so $L$ is fixed, but the speed of sound in Helium is ~3 times higher than in air. Your vocal chords produce a wide range of tones and which ones are resonant in your larynx changes when different gases are present. Similarly, sulfur hexafluoride has a speed of sound ~41% that in air so the tones that get amplified are lower.

⟹ MythBusters video